University of Central Punjab, Lahore

Department of Computer Sciences

Section: Fall 26

Assignment 1 **Submission Date: 25-10-26**

Course: Information Security **Marks: 20**

Instructions:

- Do not copy paste your assignment. In case of copy paste you will get **Zero marks.**
- ***Assignment should be in hand written form. Take clear pictures convert it into PDF format and submit it through the Portal.***
- Time management is the key of success so that you have to submit within time, in case of late your assignment will rejected and treated as zero marks.
- Each Question carry equal marks.

Q1. Convert your complete name without including a space and convert it into cipher text by using Caesar cipher with ta key of 4.

Q2. Convert the following plain text into Cipher text and vice versa by using Playfair text.

Plain Text: I am going to Karachi with security.

Key Phrase: Lahore

Q3. Using Vigenère Cipher convert the plain text into cipher and vice versa.

Plain Text: Today is very hot day

Key Phrase: Summer

Q4. Using Hill cipher technique convert the plain text into cipher and vice versa.

Plain Text: Summer

$$\mathbf{K} = \begin{pmatrix} 10 & 15 & 9 \\ 6 & 2 & 4 \\ 8 & 5 & 2 \end{pmatrix}$$

Note:

***You can assume that there is no space in the Plain Text.**